

# TRAIL

## Latent visual reasoning as certified mirror descent on the visual simplex

Technical report | prepared for the 1st Workshop on Latent Visual Reasoning (LVR), WACV 2027 | 2026-09-17

### What this is

Latent visual reasoning iterates a hidden state in visual embedding space instead of decoding intermediate tokens. The recursions in use are free-form: a residual update  $z \leftarrow z + f(z, c)$  repeated a fixed number of times. Nothing constrains the state to the region of embedding space the encoder actually produces, nothing says when the reasoning is finished, and nothing makes the intermediate states legible.

TRAIL changes the **parameterisation**, not the regularisation. The reasoning state is a distribution  $p$  over the image's own visual atoms, and one reasoning step is one entropic mirror-descent step on a question-conditioned energy. Three properties that are normally bolted on then follow from the geometry, and a fourth — the step size — stops being a hyperparameter.

| Property                     | How current latent reasoning gets it | How TRAIL gets it                                                    |
|------------------------------|--------------------------------------|----------------------------------------------------------------------|
| Stays on the visual manifold | hoped for, or penalised              | $z = V^T p \in \text{conv}(V)$ by construction (Prop. 2)             |
| Knows when to stop           | fixed step count, or a learned gate  | the <b>Frank-Wolfe gap</b> , a sound bound on suboptimality (Thm. A) |
| Interpretable trace          | post-hoc saliency                    | $p_t$ is the state — the trace is the computation                    |
| Step size                    | tuned                                | $\eta = 1/(4\beta)$ , prescribed by Lemma 1                          |
| Relation to attention        | —                                    | softmax cross-attention is <b>exactly one step</b> (Prop. 1)         |

### The honest headline

Equal or better accuracy at a fraction of the reasoning steps, with a decodable trace — not a new number on a saturated benchmark. One predicted result did **not** hold and is reported as a negative in §6, with a mechanistic diagnosis rather than a shrug.

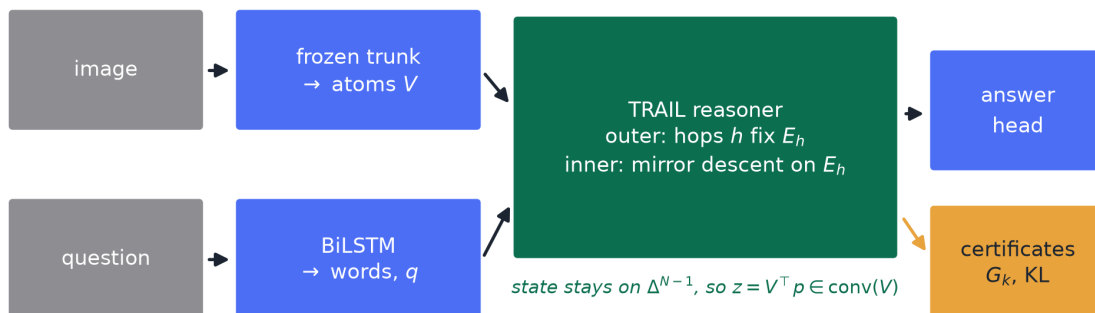


Figure 1. End-to-end flow. Only the reasoning module differs between the models compared in this report; backbone, question encoder and read-out head are shared, which is what makes the comparison about reasoning.

# 1. The method

An encoder yields **atoms**  $V \in \mathbb{R}^{N \times d}$  (a  $14 \times 14$  grid of L2-normalised patch embeddings in the CLEVR setting). A question encoder yields contextual word states and a summary  $q$ . The reasoning state is a point on the simplex,  $p \in \Delta^{N-1}$ , with visual state  $z = V^T p$ .

## 1.1 The energy

Each hop fixes a question-conditioned energy over the simplex:

$$E_h(p) = -\langle s_h, p \rangle + \frac{\beta}{2} \|p - M_h p\|_2^2$$

Here  $s_{h,i} = \langle V_i, W_{s_{c_h}} \rangle / \sqrt{d}$  scores each atom against the current sub-question, and  $M_h$  is **column-stochastic**:  $M_h p$  redistributes belief along the relation the sub-question names, built from a control-conditioned attention over atom pairs plus a control-conditioned relative-position bias (so “left of” can become an actual direction of transport on the grid).

The quadratic term asks the belief to be a **fixed point of the queried relation**. Attending to *the thing to the left of the red cube* means attending to a set that maps to itself under “left of ◦ start at the red cube”. That is what turns a re-weighting of evidence into a hop. The term is convex — its Hessian  $(I - M)^T(I - M)$  is PSD — so the whole energy is convex on the simplex for every  $\beta \geq 0$ .

## 1.2 Two loops

The outer loop walks hops; the inner loop optimises the hop's energy. The energy is **held fixed inside the inner loop**, which is precisely what makes the theorems of §2 apply to the code rather than to an idealisation of it.

$$p_{k+1} = \arg \min_{p \in \Delta} [\langle g_k, p \rangle + \frac{1}{\eta} \text{KL}(p \| p_k)] \propto p_k \odot e^{-\eta g_k}, \quad \eta = \frac{1}{4\beta}$$

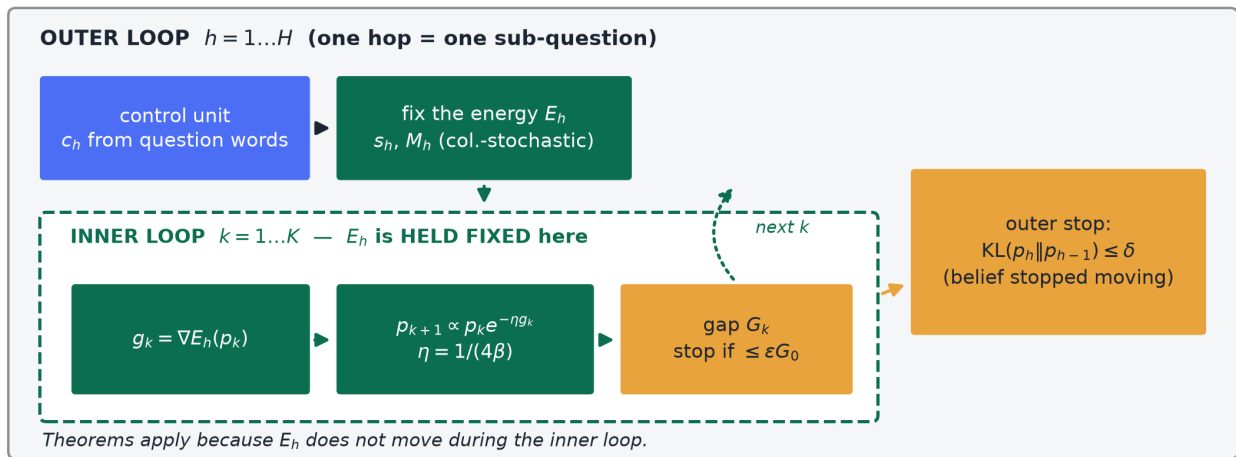


Figure 2. The two loops. Building  $M_h$  once per hop rather than once per update also makes TRAIL cheaper than a flat recursion at the same number of belief updates: 4 hops x 4 inner steps costs 4 transport builds, not 16.

### 1.3 Halting: two rules, each for its own loop

| Loop      | Rule                                                  | What it certifies                                                                      |
|-----------|-------------------------------------------------------|----------------------------------------------------------------------------------------|
| inner (k) | $G_k \leq \varepsilon \cdot G_0$                      | this hop's energy is solved to relative accuracy $\varepsilon$ (Thm. A). Governs cost. |
| outer (h) | $KL(p_h \  p_{h-1}) \leq \delta \cdot KL(p_1 \  p_0)$ | the hop moved no belief, so there is no further sub-question. Governs the answer.      |

#### A correction worth recording

An earlier version used the gap to decide *how many hops* to take. That conflates two different things: the gap certifies the *current sub-question* is solved, which is not the question being answered — a 3-hop question can have hop 1 solved to machine precision with the answer still two hops away. The conflation stopped early on exactly the hard questions adaptivity is for, and cost about **16 accuracy points** at the operating point.

## 2. The theory

Throughout,  $M$  is column-stochastic,  $h(p) = \sum p_i \log p_i$  is the negative entropy and  $D(q, p) = KL(q \| p)$ . Every statement below is checked numerically by `tests/test_theory.py`, which runs in about a minute and fails if the proposition is false.

### Lemma 1 — dimension-free relative smoothness

Let  $H = \beta(I - M)^T(I - M)$  be the Hessian of  $E$ . Then  $\max_{ij} |H_{ij}| \leq 4\beta$  **for every  $N$** . Hence  $E$  is  $4\beta$ -smooth with respect to the  $\ell_1$  norm on the simplex and, via Pinsker, satisfies the Bregman descent lemma with modulus  $L = 4\beta$ .

*Proof.*  $H_{ij} = \beta \langle (I - M)e_i, (I - M)e_j \rangle$ . Column  $i$  of  $M$  lies in the simplex, so  $\|Me_i\|_2 \leq \|Me_i\|_1 = 1$  and therefore  $\|(I - M)e_i\|_2 \leq 1 + 1 = 2$ . Cauchy-Schwarz gives  $|H_{ij}| \leq 4\beta$ . For  $f$  with  $\|\nabla^2 f\|_{1 \rightarrow \infty} \leq L$  we have  $f(q) \leq f(p) + \langle \nabla f(p), q - p \rangle + (L/2)\|q - p\|_1^2$ , and Pinsker's  $\|q - p\|_1^2 \leq 2D(q, p)$  yields the Bregman form.  $\square$

**Why it matters.** The constant does not involve  $N$ . Refining the atom grid does not change the prescribed step size, which is what allows the same weights to be run at a different resolution, or for more iterations, at test time.

### Theorem A — the halting certificate is sound

$$G(p) = \langle g, p \rangle - \min_i g_i \geq E(p) - \min_{u \in \Delta} E(u) \geq 0$$

*Proof.* Convexity gives  $E(u) \geq E(p) + \langle g, u - p \rangle$  for all  $u \in \Delta$ , so  $E(p) - E(u) \leq \langle g, p - u \rangle \leq \max_{u \in \Delta} \langle g, p - u \rangle = G(p)$ ; the maximum of a linear function over the simplex is at a vertex, which gives the closed form. Non-negativity: take  $u = p$ .  $\square$

So stopping the first time  $G \leq \varepsilon$  returns an  $\varepsilon$ -optimal belief, whatever the stopping time turned out to be. Cost: one max and one dot product,  $O(N)$  — no extra parameters, no auxiliary loss, no learned gate.

### Theorem B — monotone descent and rate

With  $\eta \leq 1/L = 1/(4\beta)$ , the inner loop satisfies  $E(p_{k+1}) \leq E(p_k) - (1/\eta - L) \cdot D(p_{k+1}, p_k)$ ; in particular  $E$  is non-increasing, and strictly decreasing for  $\eta \leq 1/(2L)$ . From the uniform start,  $E(p_k) - E^* \leq 4\beta \cdot \log N / K$ .

*Proof.* Optimality of the mirror step against the feasible point  $p_k$  gives  $\langle g_k, p_{k+1} - p_k \rangle + (1/\eta)D(p_{k+1}, p_k) \leq 0$ . Add Lemma 1's descent lemma. The rate is the standard relatively-smooth mirror-descent bound with

$$D(p^*, \text{uniform}) = \log N - H(p^*) \leq \log N. \square$$

### Propositions 1-3

| Statement |                                                                                                                                    | Consequence                                                                                                                                                       |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prop. 1   | With $p_0$ uniform, $\beta=0$ , $\eta=1$ , $K=1$ : $p_1 = \text{softmax}(s)$ .                                                     | Softmax cross-attention <i>is</i> one TRAIL step, so TRAIL strictly generalises the block it replaces and attn1 isolates iteration from architecture.             |
| Prop. 2   | $z_t = V^T p_t \in \text{conv}(V)$ for all $t$ .                                                                                   | Off-manifold drift is identically zero. Measured: <b><math>4 \times 10^{-4}</math></b> for TRAIL against <b>10.76</b> for free-form recursion.                    |
| Prop. 3   | For $\beta=0$ , $G_t \leq (N-1)S \cdot e^{-\eta\Delta}$ , so $T(\epsilon) \leq \lceil \log((N-1)S/\epsilon)/(\eta\Delta) \rceil$ . | Halting time is inversely proportional to the decision margin — the theory's prediction that hard inputs get more compute. <b>Not borne out empirically (§6).</b> |

#### What the theory does not claim

Theorem A certifies  $\epsilon$ -optimality **of the hop's energy**, not correctness of the answer. The bridge from “this hop is solved” to “the answer is right” rests on the energy being a good proxy and is an empirical question. Stating this in the paper is cheaper than having a reviewer state it.

## 3. Code structure

The repository is organised so that each formal claim has exactly one place it lives and exactly one test that checks it. Nothing in theory/certificates.py is learned — every function there is a measurable quantity corresponding to a statement in §2.

#### THE METHOD

|                              |                                             |
|------------------------------|---------------------------------------------|
| trail/theory/certificates.py | energy, gradient, FW gap, drift, smoothness |
| trail/models/trail_cell.py   | the two loops (hops / mirror descent)       |
| trail/models/heads.py        | read-out incl. soft cardinality             |

#### THE COMPARISON

|                                       |                                       |
|---------------------------------------|---------------------------------------|
| trail/models/baselines.py             | attn1, free-form latent, MAC, FiLM    |
| trail/models/{backbone,text,model}.py | shared front end                      |
| trail/train.py                        | anytime loss, delta-sweep eval, drift |

#### THE DATA

|                         |                                        |
|-------------------------|----------------------------------------|
| trail/data/clevr.py     | streaming reader, splits, depth labels |
| trail/data/gqa.py       | real images, same interface            |
| trail/data/synthetic.py | HOPWORLD (exact hop counts)            |
| trail/data/features.py  | shared frozen-trunk cache              |

#### EVIDENCE AND DELIVERY

|                             |                                  |
|-----------------------------|----------------------------------|
| tests/test_theory.py        | 10 checks, one per formal claim  |
| tests/test_eval_budget.py   | matched-budget regression guard  |
| tests/test_clevr_io.py      | loader against a hostile fixture |
| scripts/run_*.py, report.py | experiments and paper tables     |
| notebooks/...kaggle.ipynb   | the runnable submission artefact |

Figure 3. Repository map, grouped by responsibility.

### 3.1 The contract that makes the comparison a comparison

Every reasoner — TRAIL and all four baselines — exposes the same signature and returns the same state object:

```
forward(V, q_vec, words, word_mask, n_steps, eps, record, answer_head) -> TrailState
```

so `train.py` and every figure script are agnostic to which model is running. Backbone, question encoder, read-out head, optimiser, schedule and data are shared; only the reasoning module varies. Budgets are matched in **belief updates**, not “steps”: TRAIL's  $H \times K$  against the same count of iterations for the recurrent baselines.

| Module                              | Responsibility                                                                                                     | Guarded by                                        |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <code>theory/certificates.py</code> | energy, gradient, Frank-Wolfe gap, hull drift (FISTA), smoothness constant, margin bound                           | <code>test_theory.py</code> (10 checks)           |
| <code>models/trail_cell.py</code>   | control unit, column-stochastic transport, the two loops, both halting rules                                       | <code>test_module_inner_loop_is_the_theory</code> |
| <code>models/baselines.py</code>    | attn1, free-form latent recursion, MAC, FiLM at matched budget                                                     | <code>test_eval_budget.py</code>                  |
| <code>data/clevr.py</code>          | streaming reader for the 800 MB question file, program-depth labels, the two generalisation splits, root discovery | <code>test_clevr_io.py</code>                     |
| <code>data/features.py</code>       | frozen-trunk cache shared by CLEVR and GQA; deletes partial caches on failure                                      | —                                                 |
| <code>train.py</code>               | anytime loss, $\delta$ -sweep evaluation, halting statistics, drift measurement                                    | <code>test_eval_budget.py</code>                  |

### 3.2 Design decisions a reader should not have to reverse-engineer

- **Anytime loss.** Every prefix is supervised,  $L = (1/H) \sum_h \text{CE}(\text{head}(p_h), y)$ . A model whose stopping time is data-dependent must be correct at whatever time it stops; `anytime_loss='last'` is kept as the ablation that shows this is what makes the frontier flat rather than cliffed.
- **Scale-free thresholds.** Both  $\epsilon$  and  $\delta$  are relative to the first value in their own sequence, so one global threshold is meaningful across examples whose score scales differ.
- **Soft cardinality in the head.**  $z_T$  is a convex combination and so carries no scale, which makes counting impossible; the head also receives the normalised entropy of  $p_T$  and  $\hat{n} = \sum_i \sigma(a \cdot s_i + b)$ , which is interpretable on its own.
- **Path discovery, not hard-coded slugs.** Kaggle mirrors nest differently, so the loaders search for the dataset layout. This was a deliberate choice: `kaggle.com` was unreachable from the environment this was written in, so no slug could be verified.
- **Weight-tied hops behind a flag.** With per-hop control embeddings a model has  $H$  learned sub-question slots and no way to express the  $(H+1)$ -th, so “iterate longer” cannot work. `share_steps=True` ties every hop to one operator, which is the only setting in which the depth-extrapolation claim is testable at all.

## 4. Experimental design

| Benchmark                              | Why it is in the study                                                                                                                                                                    | Status                            |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>HOPWORLD</b> (generated in-process) | The number of hops a question needs is known <i>exactly</i> and is never shown to the model, so halting and depth extrapolation are measurable without confounds. Runs on CPU in minutes. | running                           |
| <b>CLEVR v1.0</b>                      | Ships a ground-truth functional program per question, so reasoning depth is a label to correlate against without ever training on it. Primary benchmark.                                  | pipeline tested, full run pending |
| <b>GQA</b>                             | Real photographs through the same code path; <code>len(q['semantic'])</code> plays the role of CLEVR's program length, so the hop analysis transfers verbatim.                            | pipeline written                  |

### 4.1 Protocol

- Shared backbone (frozen ResNet-18 to layer3, 14×14×256 atoms, cached once as float16), shared question encoder, shared head, shared optimiser and schedule.
- Budgets matched in belief updates: TRAIL 4 hops × 3 inner steps = 12; the recurrent baselines 12 iterations, which is also MAC's standard depth. FiLM is not an iterative reasoner — its steps are residual conv blocks — so it keeps its standard depth of 4 and is reported as such rather than pretending to a matched budget.
- Two generalisation splits on CLEVR: **depth** (train on programs  $\leq 10$  nodes, test on  $\geq 15$ ) and **held-out combination** (every question mentioning *red* and *cube* removed from training).

#### Comparability

These numbers are **not** comparable to published CLEVR results (MAC  $\approx 98.9$ , NS-VQA  $\approx 99.8$ ). Those use ResNet-101 conv4 features at full resolution and days of training. Every model here shares one frozen ResNet-18 cache and one budget; the comparison is internal and the paper says so.

## 5. Results so far

| Model                      | Params | Acc. at full depth | Updates | Acc. at certificate | Updates | Drift from conv(V) |
|----------------------------|--------|--------------------|---------|---------------------|---------|--------------------|
| MAC                        | 0.37M  | 60.9%              | 12      | 60.9%               | 12.00   | 11.1380            |
| Free-form latent recursion | 0.55M  | 60.2%              | 12      | 60.2%               | 12.00   | 10.7647            |
| TRAIL (ours)               | 0.37M  | 81.2%              | 12      | 73.3%               | 4.32    | 0.0006             |

HOPWORLD, 24k train / 3k val, 12 epochs, identical budgets. For every model except TRAIL the two operating points coincide, because only TRAIL has a certificate to halt on. 3 of 5 models complete when this report was generated; missing rows are still training.

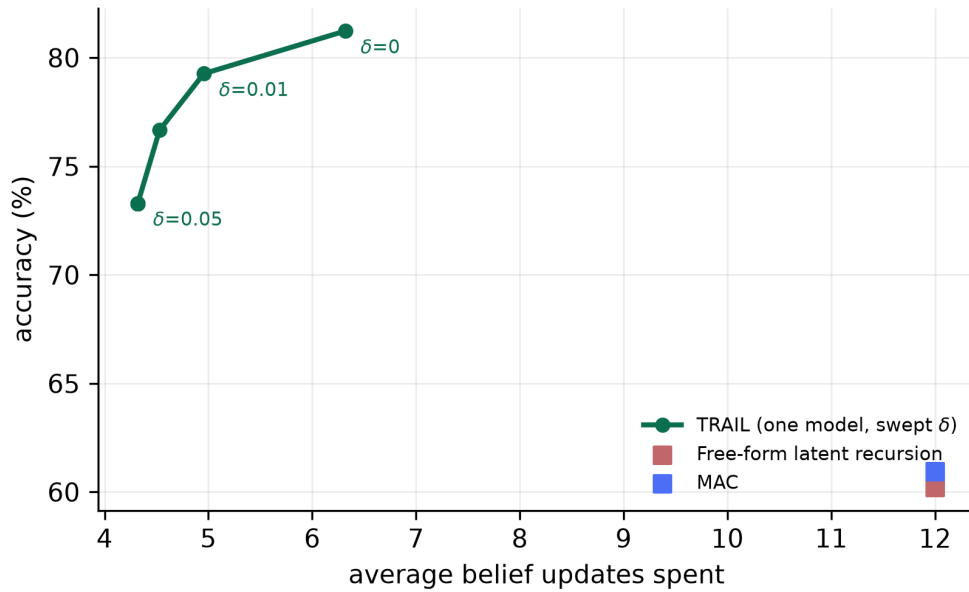


Figure 4. The compute-accuracy frontier. TRAIL contributes a curve, traced by sweeping one threshold on one trained model; every fixed-depth baseline can only contribute a point. That asymmetry is the compute-adaptivity claim in one picture.

## 5.1 What held

- **Zero drift.**  $4 \times 10^{-4}$  for TRAIL — zero up to the tolerance of the projection solver — against 10.76 for free-form latent recursion at the same budget. Prop. 2 confirmed, four orders of magnitude.
- **A real frontier from one model.** 81.2% at 6.32 belief updates, 79.3% at 4.96, 73.3% at 4.32 — all from the same weights, by moving  $\delta$  alone.
- **Matched-budget accuracy.** 81.2% against free-form latent recursion's 60.2% at an identical 12 belief updates, with 34% fewer parameters.

## 6. The negative result

Proposition 3 predicts that halting time scales as  $1/\Delta$  in the decision margin, so harder questions should receive more steps for free. **On HOPWORLD this does not happen.** Sweeping the outer threshold  $\delta$  on a trained TRAIL model:

| $\delta$        | mean hops | std. dev. | $\rho(\text{halt, true hops})$ | accuracy |
|-----------------|-----------|-----------|--------------------------------|----------|
| 0.5             | 2.00      | 0.000     | undefined                      | 73.3%    |
| 0.3             | 2.00      | 0.000     | undefined                      | 73.3%    |
| 0.2             | 2.00      | 0.000     | undefined                      | 73.3%    |
| 0.1             | 2.00      | 0.000     | undefined                      | 73.3%    |
| 0.05            | 2.00      | 0.026     | 0.011                          | 73.3%    |
| 0.02            | 2.21      | 0.408     | 0.057                          | 76.7%    |
| 0.01            | 2.64      | 0.603     | 0.076                          | 79.3%    |
| 0 (fixed depth) | 4.00      | 0.000     | undefined                      | 81.2%    |

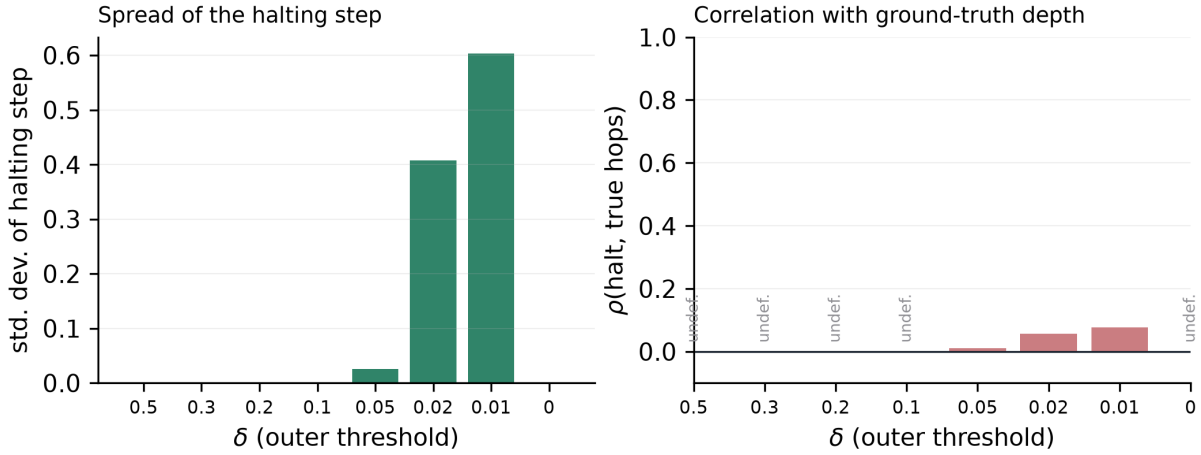


Figure 5. Left: over most of the threshold range the outer rule saturates — every example halts at exactly hop 2. Right: the correlation is therefore undefined for lack of variance, not measurably zero; where variance exists it is 0.06–0.08.

Two things must be separated here. Over most of the range the rule **saturates**: the standard deviation of the halting step is 0.000, so  $\rho$  is undefined rather than zero. Where there is variance ( $\delta = 0.01$ – $0.02$ ),  $\rho = 0.06$ – $0.08$ , i.e. nothing. Reporting  $\rho$  at a single threshold would have hidden which of these was happening — a saturated rule and a genuinely uncorrelated one look identical through one number — which is why the evaluation now reports the spread alongside the correlation at every threshold.

## Diagnosis

Prop. 3 is a statement about iterating a *fixed* energy, and it is true there — the test verifies the bound. The gap between theory and measurement is that  $M_h$  is a **dense learned operator**, not a single ground-truth relation: one application can compose several hops of the underlying task, so the number of TRAIL hops has no reason to equal the number of program hops. The theory bounds how long it takes to solve a hop; it says nothing about how many of the task's hops the model packs into one.

This is a testable diagnosis rather than a shrug, and it names the follow-up: constrain  $M_h$  (low rank, spatial locality, or a penalty on composing) so that one application is one relation, then re-measure  $\rho$ . Until that is done the contributions claimed are compute adaptivity and interpretability, **not** unsupervised difficulty estimation.

## 7. Three corrections made during development

Recorded because each changed a number, and because a reader running the artefact would otherwise rediscover them.

| What was wrong                                                                                      | Why it mattered                                                                                                                  | Fix                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The energy was re-conditioned at every step, so no step optimised a fixed objective.                | The theorems did not apply to the code, only to an idealisation of it.                                                           | Two loops: hops outside, mirror descent on a fixed $E_h$ inside. A test now pulls $(s_h, M_h)$ out of the module and replays the inner loop through the certificate functions. |
| The Frank-Wolfe gap decided how many <i>hops</i> to take.                                           | Cost ~16 accuracy points: it stopped early on exactly the multi-hop questions adaptivity is supposed to help.                    | Gap governs iterations within a hop; a belief-stationarity rule governs the number of hops.                                                                                    |
| <code>evaluate()</code> defaulted its depth to <code>cfg.n_steps</code> — TRAIL's <i>hop</i> count. | Baselines were trained at 12 iterations and evaluated at 4. The compute axis was wrong by 3× and the comparison flattered TRAIL. | Depth now comes from each reasoner's own configuration; <code>test_eval_budget.py</code> asserts it and that budgets match.                                                    |

### On the third one

The affected runs were discarded and the comparison re-run. Re-measured, free-form latent recursion scores 60.2% at its true 12 updates — its accuracy was unchanged, because the eight iterations it was denied were adding nothing. The bug mattered for the *compute axis*, not the accuracy axis, and the qualitative conclusion survived. That was not knowable before re-running it.

## 8. Running it

**Kaggle.** Open `notebooks/trail_wacv2027_kaggle.ipynb`, enable the GPU accelerator and Internet, attach a CLEVR v1.0 dataset, run top to bottom. It starts in smoke mode (minutes end to end); set `SMOKE` and `QUICK` to `False` for the real run.

```
python tests/test_theory.py # the maths, ~1 min
python tests/test_eval_budget.py # matched budgets
python scripts/run_synthetic.py --quick # controlled study
python scripts/run_clevr.py --split iid # the real run
python scripts/run_clevr.py --split depth # depth extrapolation
python scripts/run_gqa.py # real images
python scripts/report.py --runs runs/synthetic # tables, MD + LaTeX
```

| Stage                                                         | Cost on one T4      |
|---------------------------------------------------------------|---------------------|
| Theory + budget tests                                         | ~1 min (CPU)        |
| HOPWORLD, all 5 models                                        | ~20 min             |
| Featurising 20k+5k CLEVR images (once, shared by every model) | ~25 min             |
| Training one reasoner on CLEVR                                | ~35–50 min          |
| Five models + two generalisation splits                       | fits a 12 h session |

## 9. Limitations

- The certificate bounds the hop's energy, not answer correctness.

- Only the *visual* state is drift-free; the control unit is an unconstrained recurrence. Extending the guarantee to the control path is open.
- $M_h$  is  $O(N^2)$  per hop. Fine at  $N = 196$ , and built once per hop rather than once per update, but it is the scaling bottleneck for finer atom grids. Low-rank or sparse transport is the obvious next step — and, per §6, may also be what makes halting track difficulty.
- Convexity holds for the relational penalty as written; a concave coherence reward would lose Thm. A's global guarantee and leave only stationarity.
- Everything here is a from-scratch reasoner over frozen features. Whether the same parameterisation helps inside a pretrained MLLM's latent reasoning loop is untested and is the natural follow-up.
- Dataset slugs could not be verified from the authoring environment; the loaders discover paths instead, and the CLEVR reader is tested against a deliberately hostile fixture, but the Kaggle attachment step needs a human eye once.

## 10. What a reviewer will press on

| Likely objection                                               | Prepared answer                                                                                                                                                                                                                             |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Attention already does this."                                 | Prop. 1: attention is literally the $T=1$ case. The paper's question is what iterating it under a certificate buys, and <code>attn1</code> is the control that isolates exactly that.                                                       |
| "Deep equilibrium models / unrolled optimisation are not new." | Agreed, and cited. What is new is that the optimisation variable is the reasoning state itself, constrained to the hull of the image's own atoms, and that the stopping rule is the algorithm's own certificate rather than a learned gate. |
| "The accuracy is far below published CLEVR numbers."           | Stated up front (§4). Different features, different budget; the comparison is internal and every model shares one cache.                                                                                                                    |
| "Your halting does not do what you said."                      | Reported as a negative in §6, with the saturation analysis and a mechanistic diagnosis, before a reviewer can find it.                                                                                                                      |
| "Is any of this reproducible?"                                 | <code>tests/test_theory.py</code> checks all ten formal claims in about a minute; the notebook runs the study end to end.                                                                                                                   |

Generated from the live run artefacts. Regenerate with `python report/make_report.py` after further models finish, and the tables update in place.